Differential Equations Workbook

Laplace transforms

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THE LAPLACE TRANSFORM

■ 1. Find the Laplace transform, given $s > 0$.

$$\mathscr{L}(t+5)$$

■ 2. Find the Laplace transform, given $s > 0$.

$$\mathscr{L}(\sin t)$$

■ 3. Find the Laplace transform, given $s > 1$.

$$\mathscr{L}(e^t)$$

■ 4. Find the Laplace transform, given $s > 0$.

$$\mathscr{L}(5t)$$

■ 5. Find the Laplace transform, given $s > 0$.

$$\mathscr{L}(7)$$

■ 6. Find the Laplace transform, given $s > 0$.

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$\mathscr{L}(\cos t)$

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TABLE OF TRANSFORMS

■ 1. Use a table of Laplace transforms to transform the function.

$$f(t) = \cos(3t) + 5e^{-7t}$$

■ 2. Use a table of Laplace transforms to transform the function.

$$f(t) = t^2 + 3t - e^{3t}$$

■ 3. Use a table of Laplace transforms to transform the function.

$$f(t) = 2t^4 - 3\sin(2t)$$

■ 4. Use a table of Laplace transforms to transform the function.

$$f(t) = e^{-5t} - t\sin(3t)$$

■ 5. Use a table of Laplace transforms to transform the function.

$$f(t) = e^{6t}\cos t + t^4$$

■ 6. Use a table of Laplace transforms to transform the function.

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$$f(t) = \cos(5t) + 3t^3 - e^{-3t}$$

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EXPONENTIAL TYPE

■ 1. Determine the value of α in $e^{\alpha t}$ such that the function $f(t) = c$, is of exponential type. In other words, find the value of α that makes the following equation true.

$$\lim_{t\to\infty} \frac{f(t)}{e^{\alpha t}} = 0$$

■ 2. Determine the value of α in $e^{\alpha t}$ such that the function $f(t) = t$ is of exponential type. In other words, find the value of α that makes the following equation true.

$$\lim_{t\to\infty} \frac{f(t)}{e^{\alpha t}} = 0$$

■ 3. Determine the value of α in $e^{\alpha t}$ such that the function $f(t) = t^2 + t$ is of exponential type. In other words, find the value of α that makes the following equation true.

$$\lim_{t\to\infty} \frac{f(t)}{e^{\alpha t}} = 0$$

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■ 4. Determine the value of α in $e^{\alpha t}$ such that the function $f(t) = t\cos(bt)$ is of exponential type. In other words, find the value of α that makes the following equation true.

$$\lim_{t\to\infty} \frac{f(t)}{e^{\alpha t}} = 0$$

■ 5. Determine the value of α in $e^{\alpha t}$ such that the function $f(t) = \ln(t)$ is of exponential type. In other words, find the value of α that makes the following equation true.

$$\lim_{t\to\infty} \frac{f(t)}{e^{\alpha t}} = 0$$

■ 6. Determine the value of α in $e^{\alpha t}$ such that the function $f(t) = te^{bt}$ is of exponential type. In other words, find the value of α that makes the following equation true.

$$\lim_{t\to\infty} \frac{f(t)}{e^{\alpha t}} = 0$$

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PARTIAL FRACTIONS DECOMPOSITIONS

■ 1. Rewrite the function as its partial fractions decomposition.

$$f(x) = \frac{x^3 - 2x^2 + 2x - 3}{(x-2)^4}$$

■ 2. Rewrite the function as its partial fractions decomposition.

$$f(x) = \frac{x^3 + 4x^2 - 10}{x^2(x+1)(x-1)}$$

■ 3. Rewrite the function as its partial fractions decomposition.

$$f(x) = \frac{2}{(x-1)(x+1)}$$

■ 4. Rewrite the function as its partial fractions decomposition.

$$f(x) = \frac{2x^4 + 16}{x(x^2+2)^2}$$

■ 5. Rewrite the function as its partial fractions decomposition.

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$$f(x) = \frac{x+2}{(x-1)^2}$$

6. Rewrite the function as its partial fractions decomposition.

$$f(x) = \frac{x^3 + 2x^2 - 3x + 1}{x^3(x^2+1)^2}$$

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INVERSE LAPLACE TRANSFORMS

■ 1. Find the inverse Laplace transform.

$$F(s) = \frac{s+3}{s^2-3s+2}$$

■ 2. Use an inverse Laplace transform to find $f(t)$.

$$F(s) = \frac{2s-5}{s^3+s^2-12s}$$

■ 3. Find the inverse Laplace transform.

$$F(s) = \frac{s-3}{s^2+4s+4}$$

■ 4. Use an inverse Laplace transform to find $f(t)$.

$$F(s) = \frac{3s^3+6s^2+27s+24}{s^4+13s^2+36}$$

■ 5. Use an inverse Laplace transform to find $f(t)$.

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$$F(s) = \frac{s-2}{s^2-4s+5}$$

6. Find the inverse Laplace transform.

$$F(s) = \frac{s-2}{2s^2+2s+2}$$

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TRANSFORMING DERIVATIVES

- 1. Find the Laplace transforms of $y'(t)$ and $y''(t)$, given $y(0) = 1$ and $y'(0) = 1$.

- 2. Find the Laplace transforms of $y'(t)$ and $y''(t)$, given $y(0) = 0$ and $y'(0) = 2$.

- 3. Find the Laplace transforms of $y'(t)$ and $y''(t)$, given $y(0) = 1$ and $y'(0) = -2$.

- 4. Find the Laplace transforms of $y'(t)$ and $y''(t)$, given $y(0) = 3$ and $y'(0) = -1/4$.

- 5. Find the Laplace transforms of $y'(t)$ and $y''(t)$, given $y(0) = 4$ and $y'(0) = 2/3$.

- 6. Find the Laplace transforms of $y'(t)$ and $y''(t)$, given $y(0) = 7$ and $y'(0) = 1/7$.

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LAPLACE TRANSFORMS FOR INITIAL VALUE PROBLEMS

■ 1. Find the solution to the second order equation, given $y(0) = 1$ and $y'(0) = -1$.

$$y'' + 3y' + 2y = 6t$$

■ 2. Find the solution to the second order equation, given $y(0) = 1$ and $y'(0) = -1$.

$$3y'' + 4y' + y = -t$$

■ 3. Find the solution to the second order equation, given $y(0) = 1$ and $y'(0) = 4$.

$$y'' - 5y' = 10t - 2$$

■ 4. Find the solution to the second order equation, given $y(0) = 2$ and $y'(0) = 2$.

$$y'' - 4y = -13\cos(3t)$$

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■ 5. Find the solution to the second order equation, given $y(2) = 1$ and $y'(2) = -2$.

$$y'' + 2y' - 3y = 0$$

■ 6. Find the solution to the second order equation, given $y(\pi) = 2$ and $y'(\pi) = 0$.

$$y'' + 4y = 0$$

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STEP FUNCTIONS

■ 1. Write $f(t)$ in terms of step functions, if $f(t)$ has a value of -3 at $t = 0$, jumps down 2 units at $t = 4$, up 7 units at $t = 5$, and if $f(t) = 9$ for $t \geq 6$.

■ 2. Write an (on at 5)-to-off function that has a switch at $t = 8$.

■ 3. Express the piecewise function in terms of unit step functions.

$$f(t) = \begin{cases} \cos t & 0 \leq t < 2\pi \\ 0 & t \geq 2\pi \end{cases}$$

■ 4. Rewrite the function $g(t)$ in terms of Heaviside functions.

$$g(t) = \begin{cases} 0 & 0 \leq t < 1 \\ t^2 - 3 & 1 \leq t < 2 \\ 5 & 2 \leq t < 3 \\ t^3 + 2 & t \geq 3 \end{cases}$$

■ 5. Describe all switches of the function $f(t) = (-5 + 6u_2(t) - 3u_5(t))(1 - u_7(t))$ and represent it as a piecewise function.

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■ 6. Describe all switches of the function $f(t) = 4 - u_1(t) + 5u_3(t) + 3u_4(t) - 6u_8(t)$ and represent it as a piecewise function.

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SECOND SHIFTING THEOREM

■ 1. Use a step function to represent shifting the portion of $f(x)$ for $x \geq 0$ to the right 2 units, while turning off the function on $0 \leq x < 2$. Then take its Laplace transform.

$$f(x) = 5 - x^2 + \cos(3x) - 4e^{5x}$$

■ 2. Use a step function to represent shifting the portion of $f(x)$ for $x \geq 0$ to the right 3 units, while turning off the function on $0 \leq x < 3$. Then take its Laplace transform.

$$f(x) = 7x \sin x - 4\cos(3x + 9)$$

■ 3. Find the Laplace transform of the function.

$$g(t) = \begin{cases} 0 & 0 \leq x < 7 \\ e^{3t-21} & x \geq 7 \end{cases}$$

■ 4. Find the Laplace transform of the function.

$$g(t) = \begin{cases} 0 & 0 \leq x < 4 \\ e^{t-4}\sin(3t - 12) - (t-4)^3 e^{4t-16} & x \geq 4 \end{cases}$$

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■ 5. Calculate the Laplace transform.

$$\mathscr{L}(e^{3t-6}\cos(4-2t)u(t-2))$$

■ 6. Find the Laplace transform of the function.

$$f(t)=\begin{cases}1-t & 0\le x<2\\ -1 & 2\le x<4\\ (5t-20)^3e^{3t-12} & x\ge 4\end{cases}$$

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LAPLACE TRANSFORMS OF STEP FUNCTIONS

■ 1. Find the inverse Laplace transform.

$$F(s) = \frac{3se^{-7s}}{s^2 - 5s + 6}$$

■ 2. Find the inverse Laplace transform.

$$G(s) = \frac{(2s^2 + 1)e^{-3s}}{s^3 - 4s^2 + 3s}$$

■ 3. Find the Laplace transform of the function $g(t)$.

$$g(t) = 5u_1(t) + 6(t-3)^3 u_3(t) + 7e^4 u_4(t)$$

■ 4. Find the Laplace transform of the function $g(t)$.

$$f(t) = \begin{cases} 3 & 0 \le t < 2 \\ t + 1 & 2 \le t < 5 \\ 4t - 14 & t \ge 5 \end{cases}$$

■ 5. Find the Laplace transform of the function $g(t)$.

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$$f(t) = \begin{cases} 0 & 0 \leq t < \frac{\pi}{2} \\ 3\sin t & \frac{\pi}{2} \leq t < \pi \\ \cos t & t \geq \pi \end{cases}$$

6. Find the inverse Laplace transform.

$$G(s) = \frac{3}{s} + e^{-s}\left(\frac{1}{s} - \frac{2}{s-4}\right) + e^{-3s}\frac{1}{s-9}$$

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STEP FUNCTIONS WITH INITIAL VALUE PROBLEMS

■ 1. Solve the initial value problem, given $y(0) = 0$ and $y'(0) = 0$.

$$y'' = \begin{cases} 2 & 0 \le t < 4 \\ e^{2t} & t \ge 4 \end{cases}$$

■ 2. Solve the initial value problem, given $y(0) = 1$ and $y'(0) = 0$.

$$y'' - 9y = e^{2t-10}u(t-5)$$

■ 3. Solve the initial value problem, given $y(0) = -1$ and $y'(0) = 1$.

$$y'' - 4y' = g(t)$$

$$g(t) = \begin{cases} 0 & 0 \le t < 5 \\ -3 & t \ge 5 \end{cases}$$

■ 4. Solve the initial value problem, given $y(0) = 0$ and $y'(0) = 0$.

$$y'' - 5y' + 6y = tu(t-4)$$

■ 5. Solve the initial value problem, given $y(0) = -2$ and $y'(0) = 1$.

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$$y'' - y' = \sin(3t - 9)u(t - 3)$$

■ 6. Solve the initial value problem, given $y(0) = 0$ and $y'(0) = 2$.

$$y'' - 6y' = \begin{cases} 1 & 0 \le t < 2 \\ 3t + 1 & t \ge 2 \end{cases}$$

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THE DIRAC DELTA FUNCTION

■ 1. Solve the initial value problem, given $y(0) = 1$ and $y'(0) = 0$.

$$y'' + 4y = 4\delta(t - 2)$$

■ 2. Solve the initial value problem, given $y(0) = 0$ and $y'(0) = 1$.

$$y'' + 2y' + 3y = \delta(t - 5)$$

■ 3. Solve the initial value problem, given $y(0) = 2$ and $y'(0) = 0$.

$$y'' - y' = 3u(t - 1)$$

■ 4. Solve the initial value problem, given $y(0) = 1$ and $y'(0) = 1$.

$$y'' + y' - 2y = 3u(t - 4)$$

■ 5. Solve the initial value problem, given $y(0) = -2$ and $y'(0) = 1$.

$$y'' - 2y = u(t - 7) - 2\delta(t - 7)$$

■ 6. Solve the initial value problem, given $y(0) = -1$ and $y'(0) = -2$.

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$$y'' + 2y' + y = 7\delta(t-5) + 5u(t-3)$$

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CONVOLUTION INTEGRALS

■ 1. Find the convolution of $f(t) = e^{2t}$ and $g(t) = 2t$, then show that the Laplace transform of the convolution is equivalent to the product of the individual transforms $F(s)$ and $G(s)$.

■ 2. Use a convolution integral to find the inverse transform.

$$H(s) = \frac{s^2}{(s^2+1)^2}$$

■ 3. Find the convolution of $f(t) = \cos t$ and $g(t) = t^2$, then show that the Laplace transform of the convolution is equivalent to the product of the individual transforms $F(s)$ and $G(s)$.

■ 4. Find the convolution of $f(t) = e^{2t}$ and $g(t) = e^{-3t}$, then show that the Laplace transform of the convolution is equivalent to the product of the individual transforms $F(s)$ and $G(s)$.

■ 5. Use a convolution integral to find the inverse transform of the following transform.

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$$H(s) = \frac{1}{(s-5)(s-6)}$$

■ 6. Find the convolution of $f(t) = t^3$ and $g(t) = 8$, then show that the Laplace transform of the convolution is equivalent to the product of the individual transforms $F(s)$ and $G(s)$.

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CONVOLUTION INTEGRALS FOR INITIAL VALUE PROBLEMS

■ 1. Use a convolution integral to find the general solution $y(t)$ to the differential equation, given $y(0) = 0$ and $y'(0) = 0$.

$$y'' + 4y = g(t)$$

■ 2. Use a convolution integral to find the general solution $y(t)$ to the differential equation, given $y(0) = 0$ and $y'(0) = -2$.

$$y'' + 2y' - 3y = g(t)$$

■ 3. Use a convolution integral to find the general solution $y(t)$ to the differential equation, given $y(0) = 1$ and $y'(0) = 0$.

$$y'' - y' - 2y = e^{5t}$$

■ 4. Use a convolution integral to find the general solution $y(t)$ to the differential equation, given $y(0) = 2$ and $y'(0) = -1$.

$$y'' + 3y' = g(t)$$

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■ 5. Use a convolution integral to find the general solution $y(t)$ to the differential equation, given $y(0) = 1$ and $y'(0) = 1$.

$$y'' - 5y' + 4y = g(t)$$

■ 6. Use a convolution integral to find the general solution $y(t)$ to the differential equation, given $y(0) = -1$ and $y'(0) = 0$.

$$y'' - 2y' - 8y = g(t)$$

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